Ola Data Analysis Project

The data will be for 1 month. use the following column –

1. Date
2. Time
3. Booking ID
4. Booking Status
5. Customer ID
6. Vehicle Type –
    - Auto - Prime Plus - Prime Sedan - Mini - Bike - eBike - Prime SUV
7. Pickup Location
8. Drop Location (Take from dummy pickup locations)
9. Avg VTAT (Time taken to arrive at the vehicle)
10. Avg CTAT (Time taken to arrive the Customer)
11. Cancelled Rides by Customer
12. Reason for cancelling by Customer –
    - Driver is not moving towards pickup location - Driver asked to cancel - AC is not working (Only for 4-wheelers) - Change of plans - Wrong Address
13. Cancelled Rides by Driver –
    - Personal & Car related issues - Customer related issue - The customer was coughing/sick - More than permitted people in there
14. Incomplete Rides
15. Incomplete Rides Reason - Customer Demand - Vehicle Breakdown - Other Issue
16. Booking Value

17. Ride Distance
18. Driver Ratings
19. Customer Rating

SQL Questions:

1. Retrieve all successful bookings:
2. Find the average ride distance for each vehicle type:
3. Get the total number of cancelled rides by customers:
4. List the top 5 customers who booked the highest number of rides:
5. Get the number of rides cancelled by drivers due to personal and car-related issues:
6. Find the maximum and minimum driver ratings for Prime Sedan bookings:
7. Retrieve all rides where payment was made using UPI:
8. Find the average customer rating per vehicle type:
9. Calculate the total booking value of rides completed successfully:
10. List all incomplete rides along with the reason:

Power BI Questions:

1. Ride Volume Over Time
2. Booking Status Breakdown
3. Top 5 Vehicle Types by Ride Distance
4. Average Customer Ratings by Vehicle Type
5. cancelled Rides Reasons
6. Revenue by Payment Method
7. Top 5 Customers by Total Booking Value
8. Ride Distance Distribution Per Day

9. Driver Ratings Distribution

10. Customer vs. Driver Ratings

SQL Analysis Questions and Answers

```
Create Database Ola;
use Ola;
select * from Bookings ;
```

-- 1. Retrieve all successful bookings:

```
Create View Successful_Bookings as
select * from Bookings
Where Booking_Status = 'Success' ;

select * from Successful_Bookings ;
```

-----------------x----------------------------x-------------------------x-----------

-- 2. Find the average ride distance for each vehicle type:

```
CREATE VIEW Avg_Ride_Distance_by_Vehicle AS
select Vehicle_Type , AVG(Ride_Distance) as Avg_Distance
from Bookings group by Vehicle_Type ;

select * from Avg_Ride_Distance_by_Vehicle ;
```

-----------------x----------------------------x-------------------------x-----------

-- 3. Get the total number of cancelled rides by customers:

```
Create View Total_Cancelled_By_Customers AS
select count(*) as Total_Cancelled from Bookings
where Booking_Status = 'Canceled by Customer' ;

Select * from Total_Cancelled_By_Customers ;
```

-----------------x----------------------------x-------------------------x-----------

-- 4. List the top 5 customers who booked the highest number of rides:

```
Create View Top_5_Customers as
select Top 5 Customer_ID , count(Booking_ID) as Total_Rides
from Bookings Group by Customer_ID Order by Total_Rides DESC ;

select * from Top_5_Customers ;
```

-----------------x----------------------------x-------------------------x-----------

-- 5. Get the number of rides cancelled by drivers due to personal and car-related issues:

```
Create View Rides_Canceled_by_Drivers_P_C_Issues as
select count(*) as Total_Rides from Bookings
Where Canceled_Rides_by_Driver = 'Personal & Car related issue' ; --- 4449
```

select * from Rides_Canceled_by_Drivers_P_C_Issues ;

-----------------x----------------------------x-------------------------x-----------

-- 6. Find the maximum and minimum driver ratings for Prime Sedan bookings:

```
Create View Max_Min_Driver_Rating as
Select
Max(Driver_Ratings) as Maximum_Rating,
Min(Driver_Ratings) as Minimum_Rating
from Bookings Where Vehicle_Type = 'Prime Sedan' ;
```

-----------------x----------------------------x-------------------------x-----------

-- 7. Retrieve all rides where payment was made using UPI:

```
Create View All_UPI_Payment as
select * from Bookings
Where Payment_Method = 'UPI' ;

select * from All_UPI_Payment ;
```

-----------------x----------------------------x-------------------------x-----------

-- 8. Find the average customer rating per vehicle type:

```
Create View Avg_Customer_Rating_By_VehicleType as
select Vehicle_Type , CAST(AVG(Customer_Rating) as decimal(10,2)) as Average_Rating
from Bookings Group by Vehicle_Type ;

select * from Avg_Customer_Rating_By_VehicleType ;
```

-----------------x----------------------------x-------------------------x-----------

-- 9. Calculate the total booking value of rides completed successfully:

```
Create View Total_Booking_Value_of_Success_Rides as
select sum(Booking_Value) as Total_Booking_Value
from Bookings Where Booking_Status = 'Success' ;

select * from Total_Booking_Value_of_Success_Rides
```

-----------------x----------------------------x-------------------------x-----------

-- 10. List all incomplete rides along with the reason:

```
Create View Incomplete_Rides as
select Booking_ID , Incomplete_Rides , Incomplete_Rides_Reason
from Bookings where Incomplete_Rides = 'Yes'

select * from Incomplete_Rides ;
```

-----------------x----------------------------x-------------------------x-----------

All Results

select * from Bookings ;

-- 1. Retrieve all successful bookings:

select * from Successful_Bookings ;

-----------------x---------------------------x-------------------------x-----------

-- 2. Find the average ride distance for each vehicle type:

select * from Avg_Ride_Distance_by_Vehicle ;

-----------------x---------------------------x-------------------------x-----------

-- 3. Get the total number of cancelled rides by customers:

Select * from Total_Cancelled_By_Customers ;

-----------------x---------------------------x-------------------------x-----------

-- 4. List the top 5 customers who booked the highest number of rides:

select * from Top_5_Customers ;

-----------------x---------------------------x------------------------x-----------

-- 5. Get the number of rides cancelled by drivers due to personal and car-related issues:

select * from Rides_Canceled_by_Drivers_P_C_Issues ;

-----------------x---------------------------x------------------------x-----------

-- 6. Find the maximum and minimum driver ratings for Prime Sedan bookings:

select * from Max_Min_Driver_Rating ;

-----------------x---------------------------x------------------------x-----------

-- 7. Retrieve all rides where payment was made using UPI:

select * from All_UPI_Payment ;

-----------------x---------------------------x------------------------x-----------

-- 8. Find the average customer rating per vehicle type:

select * from Avg_Customer_Rating_By_VehicleType ;

-----------------x---------------------------x------------------------x-----------

-- 9. Calculate the total booking value of rides completed successfully:

select * from Total_Booking_Value_of_Success_Rides

-----------------x----------------------------x-------------------------x-----------

-- 10. List all incomplete rides along with the reason:

select * from Incomplete_Rides ;

-----------------x----------------------------x-------------------------x-----------